

# Autonomous Control of Quadcopter–Pendulum System for Induced Oscillations and Stabilization

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ME494 : B. Tech. Project

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# INTRODUCTION

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- Unmanned Aerial Vehicles (UAVs) are increasingly used for aerial delivery, inspection, and construction.
- Carrying suspended payloads is a challenging configuration due to significant nonlinearities and underactuated dynamics.
- Payload swing can destabilise the flight controller and degrade tracking performance.



**Primary Aim:** Utilising the dynamics of a suspended load to propel a payload into a target zone.

- The drone autonomously "energises" the system, so no need for multiple landings.
- Allows delivery to locations the drone cannot physically enter, increasing the effective range of the drone
- Keeps the expensive aerial platform at a safe standoff distance from obstacles

# OBJECTIVES

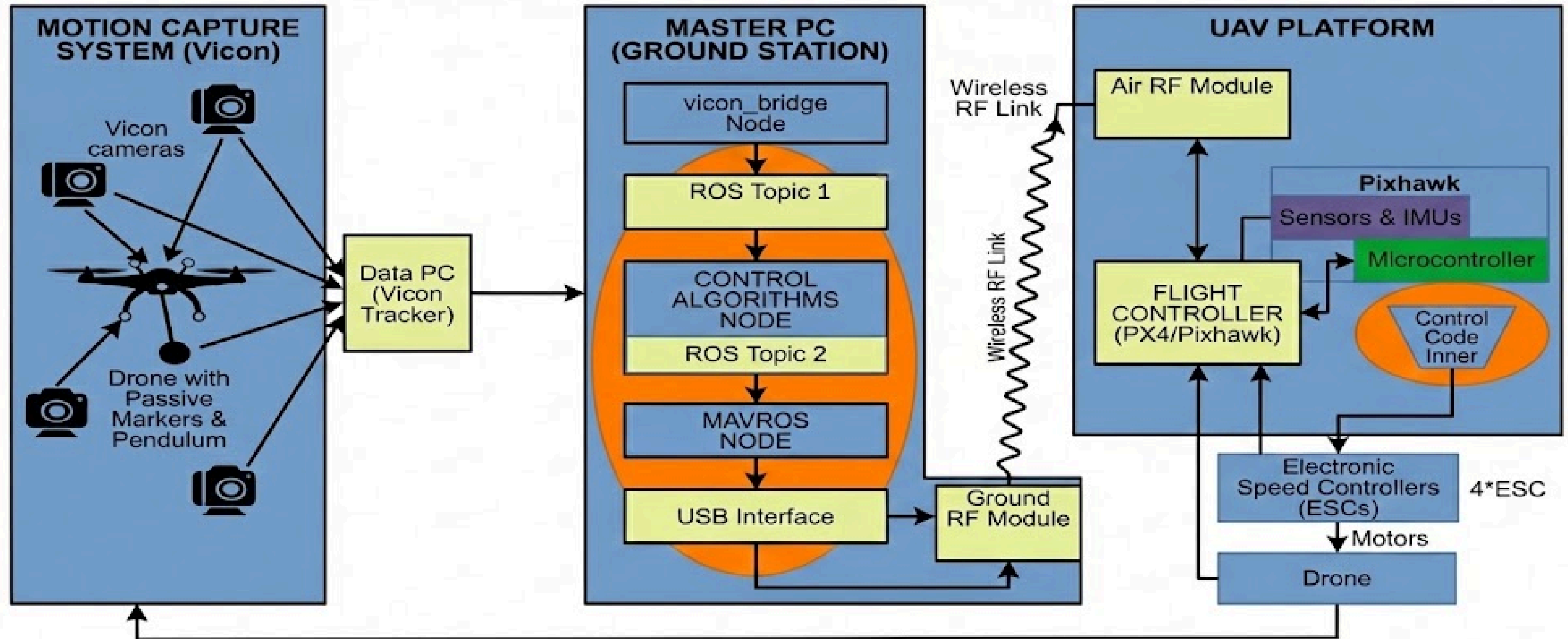
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The primary aim of this work is to design an autonomous control system for a quadcopter-pendulum system using ROS.

The key goals of this project are:

- To implement a real-time, closed-loop control framework interfacing with the PX4 flight controller via MAVROS.
- Design a strategy to intentionally induce and sustain stable oscillations to a target amplitude.
- Integrate an electromagnetic locker and autonomous logic to secure the pendulum at a specific angle.
- Develop an active damping controller to rapidly stabilise the system after the manoeuvre.

# CONTROL LOOP

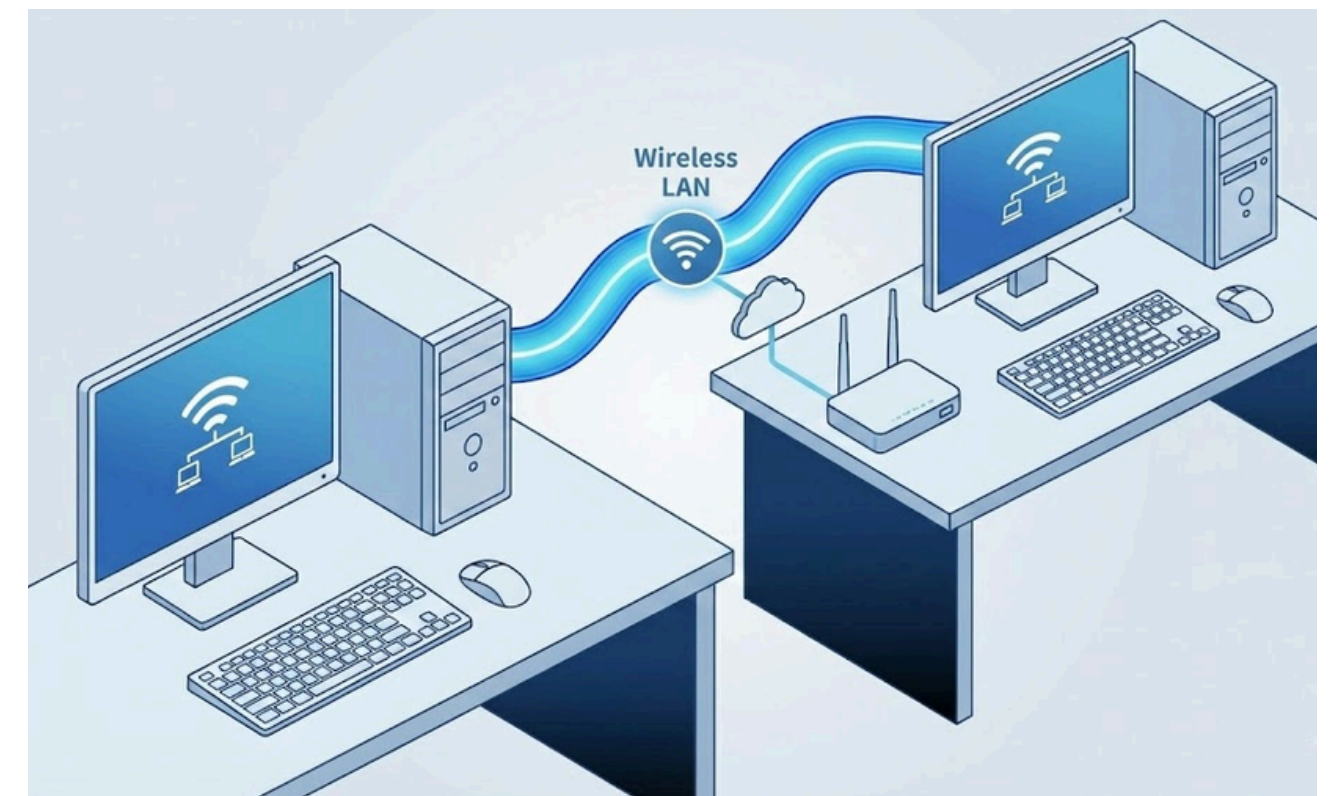


# DATA TRANSPORT ARCHITECTURE

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## DATA PC-MASTER PC:

- Both the Data PC (State Estimation) and Master PC (Control) connect to a common Local Area Network (LAN) via a Wi-Fi router.
- Communication uses the distributed networking capabilities of ROS over an IP layer.
- ROS\_MASTER\_URI: Configured on both machines to point to the Master PC's IP address, locating the roscore service.
- ROS\_IP: Each machine advertises its own local IP address to ensure bidirectional visibility.
- Machines export their network configuration to the Master to negotiate connections.
- Data packets are transmitted directly between nodes.
- This setup is susceptible to signal interference and variable packet delays

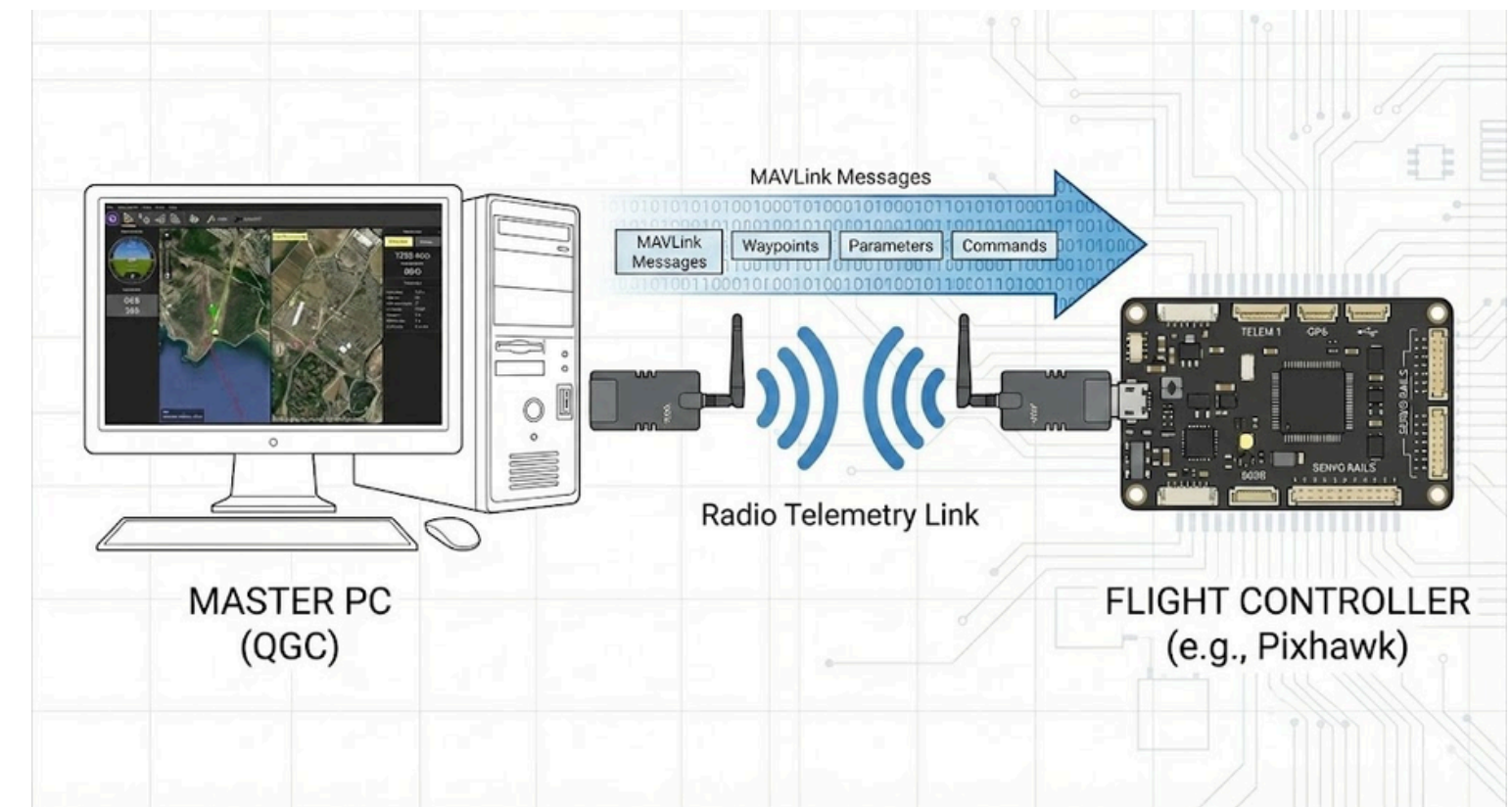


# DATA TRANSPORT ARCHITECTURE

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## MASTER PC - FLIGHT CONTROLLER:

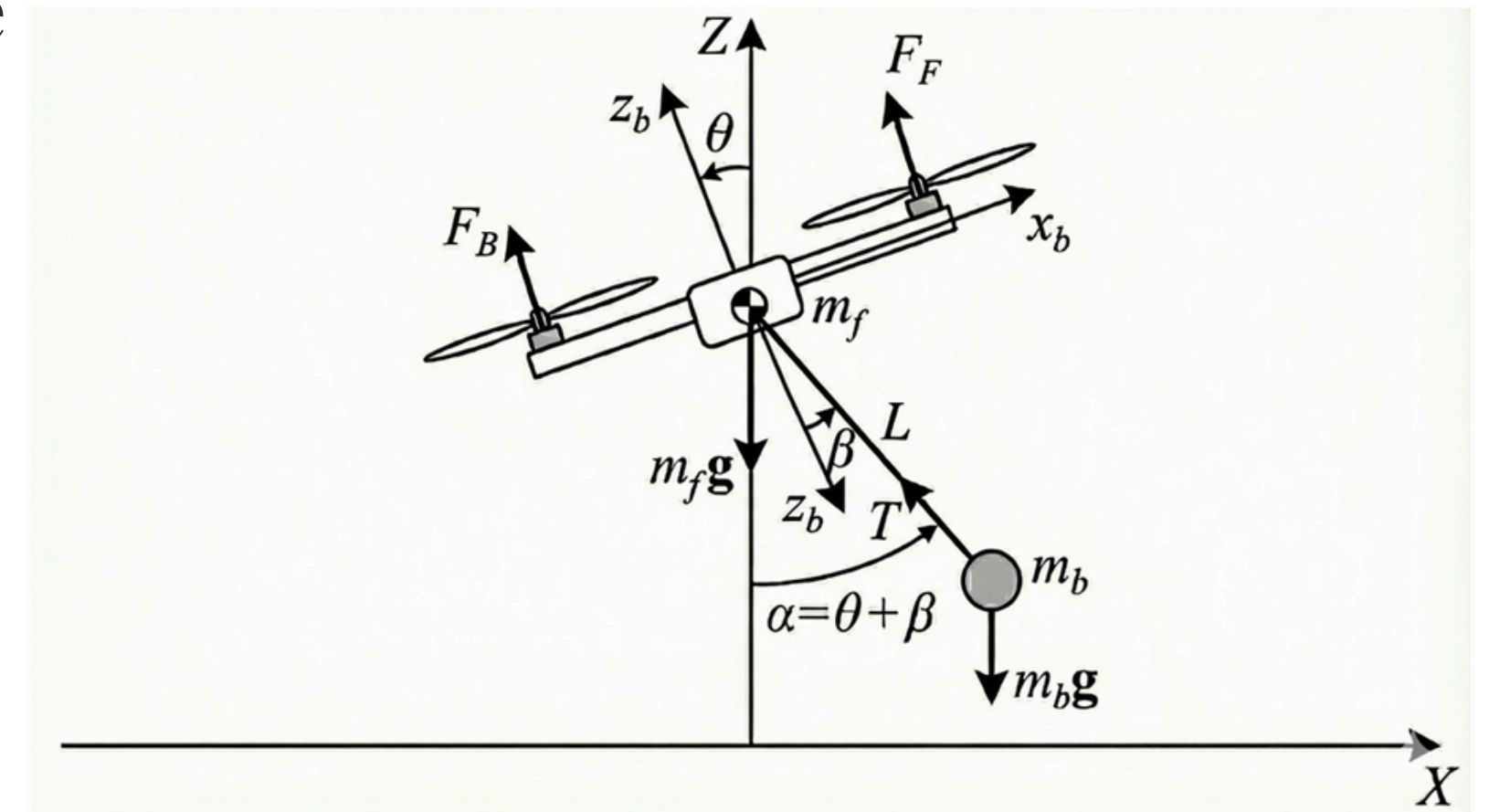
- The link between the high-level processor (Master PC) and low-level control (Drone) is established via a dedicated Radio Frequency (RF) pair.
  - Ground Module: Connected to the Master PC via USB.
  - Air Module: Interfaced with the drone's PX4 Flight Controller via a Telemetry (UART) port.
- MAVROS: A ROS node running on the PC converts control commands into MAVLink packets.
- The pre-paired RF modules create a wireless serial tunnel, enabling bi-directional communication.
- Uplink (Commands): Transmits Offboard setpoints (x,y, z, roll, pitch, yaw ) and configuration parameters.
- Downlink (Telemetry): Receives real-time sensor data (IMU, Battery) and arming status.



# SIMULATION

## System Definition & Assumptions:

- 2D Planar Dynamics: Motion is restricted to the vertical longitudinal plane (x-z), assuming yaw is aligned with the target
- The state is defined by  $\{q\} = [x, z, \theta, \beta]$ 
  - x, z: Drone Position;  $\theta$ : Drone Pitch;  $\beta$ : Pendulum Angle relative to body.
- $X, Z$  : Global x and z coordinates
- i,k: unit vectors wrt drone body
- $U_1$ : Total Thrust (Lift).
- $U_2$ : Pitching Moment (Torque).



$$U_1 = F_F + F_B$$

$$U_2 = (F_F - F_B) \cdot d$$

# SIMULATION

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## Swing-Up Dynamics:

For the excitation phase, we decouple the system into the quadcopter rigid body dynamics and the relative pendulum dynamics.

### Drone Body:

$$m_f \ddot{x} = -(F_F + F_B) \sin \theta + T \sin(\theta + \beta)$$

$$m_f \ddot{z} = (F_F + F_B) \cos \theta - m_f g - T \cos(\theta + \beta)$$

$$I_y \ddot{\theta} = (F_F - F_B) d = U_2$$

# SIMULATION

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## Swing-Up Dynamics:

### Pendulum relative to drone:

Since we are considering dynamics relative to the body, we must consider all the pseudo-forces.

**Gravity**  $(F_g) = -m_b g \hat{\mathbf{Z}} = -m_b g [\sin(\theta) \hat{\mathbf{i}} + \cos(\theta) \hat{\mathbf{k}}]$

**Euler Forces**  $F_{\text{Euler}} = m_b L \ddot{\theta} [\cos(\beta) \hat{\mathbf{i}} + \sin(\beta) \hat{\mathbf{k}}]$

**Tension in the link**  $(T) = -T [\sin(\beta) \hat{\mathbf{i}} - \cos(\beta) \hat{\mathbf{k}}]$

**Centrifugal Forces**  $F_{\text{Cent}} = m_b L \dot{\theta}^2 [\sin(\beta) \hat{\mathbf{i}} - \cos(\beta) \hat{\mathbf{k}}]$

**Translational Pseudo Forces**  $F_{\text{trans}} = -m_b \ddot{x} \mathbf{X} - m_b \ddot{z} \mathbf{Z}$   
$$= -[m_b \ddot{x} \cos(\theta) + m_b \ddot{z} \sin(\theta)] \hat{\mathbf{i}}$$
$$+ [m_b \ddot{x} \sin(\theta) - m_b \ddot{z} \cos(\theta)] \hat{\mathbf{k}}$$

**Coriolis Forces**  $F_{\text{Cor}} = -2m_b L \dot{\theta} \dot{\beta} [\sin(\beta) \hat{\mathbf{i}} - \cos(\beta) \hat{\mathbf{k}}]$

# SIMULATION

## Swing-Up Dynamics:

### Pendulum relative to drone:

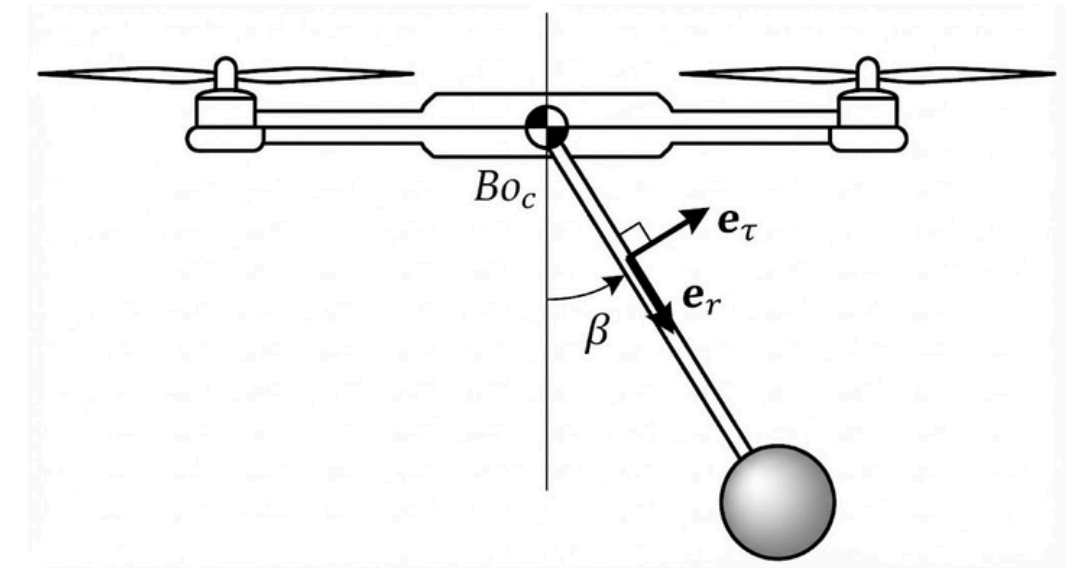
- Taking the dot product of the total force  $F_{\text{total}}$  with the tangential vector  $e_\tau$  yields the effective force driving the angular acceleration:

$$m_b L \ddot{\beta} = m_b L \ddot{\theta} - m_b g \sin(\theta + \beta) - m_b \ddot{x} \cos(\theta + \beta) - m_b \ddot{z} \sin(\theta + \beta)$$

$$\ddot{\beta} = \ddot{\theta} - \frac{1}{L} [g \sin(\theta + \beta) + \ddot{x} \cos(\theta + \beta) + \ddot{z} \sin(\theta + \beta)]$$

- Similarly, in radial direction it yields tension in the link

$$T = m_b L (\dot{\theta} - \dot{\beta})^2 + m_b g \cos(\theta + \beta) + m_b [\ddot{z} \cos(\theta + \beta) - \ddot{x} \sin(\theta + \beta)]$$



$$\hat{e}_r = -\sin(\beta)\hat{i} + \cos(\beta)\hat{k}$$

$$\hat{e}_\tau = \cos(\beta)\hat{i} + \sin(\beta)\hat{k}$$

# SIMULATION

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## Swing-Up Dynamics:

Substituting in the equations:

$$\mathbf{M} \begin{bmatrix} \ddot{x} \\ \ddot{z} \end{bmatrix} = \mathbf{C} \quad \mathbf{M} = \begin{bmatrix} m_f + m_b \sin^2(\gamma) & -m_b \sin(\gamma) \cos(\gamma) \\ -m_b \sin(\gamma) \cos(\gamma) & m_f + m_b \cos^2(\gamma) \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} -U_1 \sin \theta + m_b g \sin(\gamma) \cos(\gamma) + m_b L (\dot{\beta} + \dot{\theta})^2 \sin(\gamma) \\ U_1 \cos \theta - m_f g - m_b g \cos^2(\gamma) - m_b L (\dot{\beta} + \dot{\theta})^2 \cos(\gamma) \end{bmatrix}$$

$$\ddot{\beta} = \ddot{\theta} - \frac{1}{L} [g \sin(\theta + \beta) + \ddot{x} \cos(\theta + \beta) + \ddot{z} \sin(\theta + \beta)]$$

$$\ddot{\theta} = \frac{U_2}{I_y}$$

# SIMULATION

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## Swing-Up Dynamics:

The simulation advances the system state in discrete time steps using the Forward Euler integration method.

$$\begin{aligned}x_{k+1} &= x_k + \dot{x}_k \Delta t, & \dot{x}_{k+1} &= \dot{x}_k + \ddot{x} \Delta t \\z_{k+1} &= z_k + \dot{z}_k \Delta t, & \dot{z}_{k+1} &= \dot{z}_k + \ddot{z} \Delta t \\\theta_{k+1} &= \theta_k + \dot{\theta}_k \Delta t, & \dot{\theta}_{k+1} &= \dot{\theta}_k + \ddot{\theta} \Delta t \\\beta_{k+1} &= \beta_k + \dot{\beta}_k \Delta t, & \dot{\beta}_{k+1} &= \dot{\beta}_k + \ddot{\beta} \Delta t\end{aligned}$$

# SIMULATION

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## Control Logic:

### Altitude Control:

$$a_{z,cmd} = K_{p,z}(z_{des} - z) + K_{d,z}(\dot{z}_{des} - \dot{z})$$

$$U_1 = \frac{(m_f + m_b)(g + a_{z,cmd})}{\cos \theta}$$

### Swing-up Control:

We utilise an energy-based control law that modulates the drone's pitch angle  $\theta$  to excite the pendulum's natural frequency

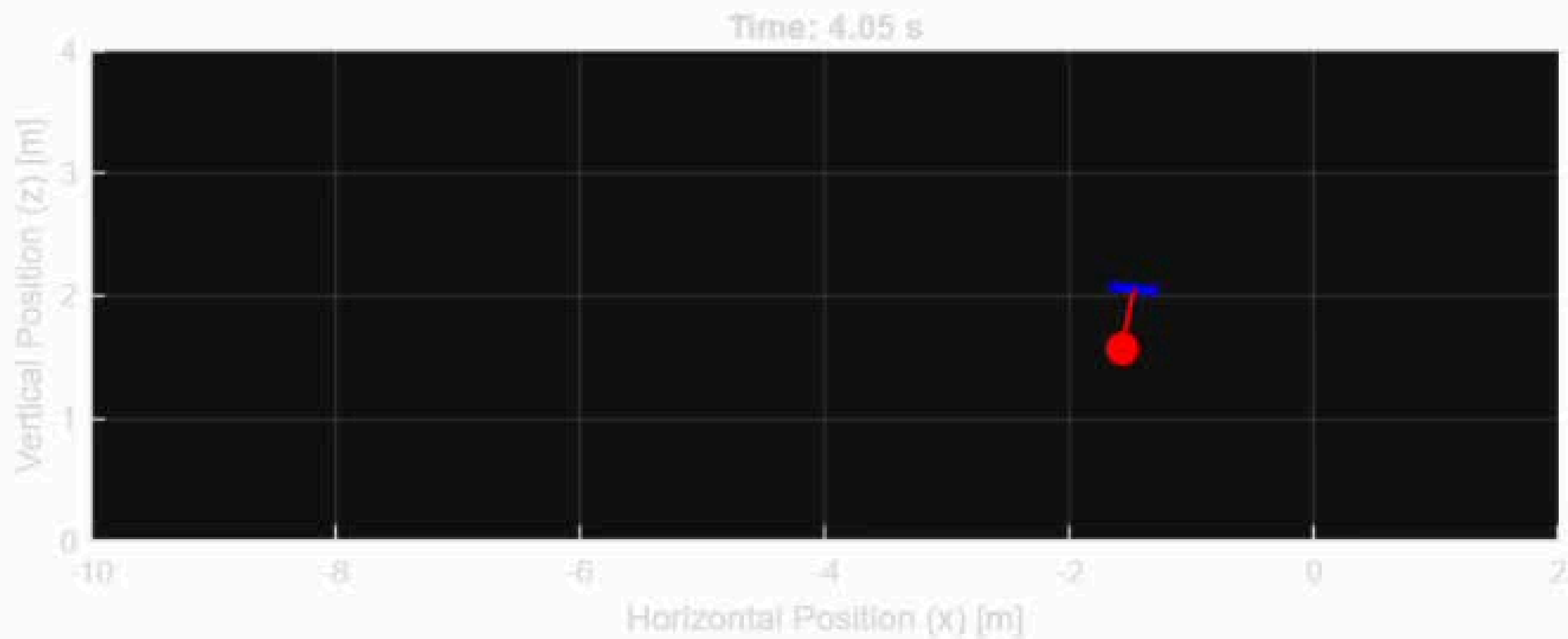
$$E(\beta, \dot{\beta}) = \frac{1}{2}m_b L^2 \dot{\beta}^2 + m_b g L(1 - \cos \beta)$$

$$\theta_{des} = \begin{cases} -K_{swing} \cdot \dot{\beta} & \text{if } E < E_{target} \\ 0 & \text{if } E \geq E_{target} \end{cases}$$

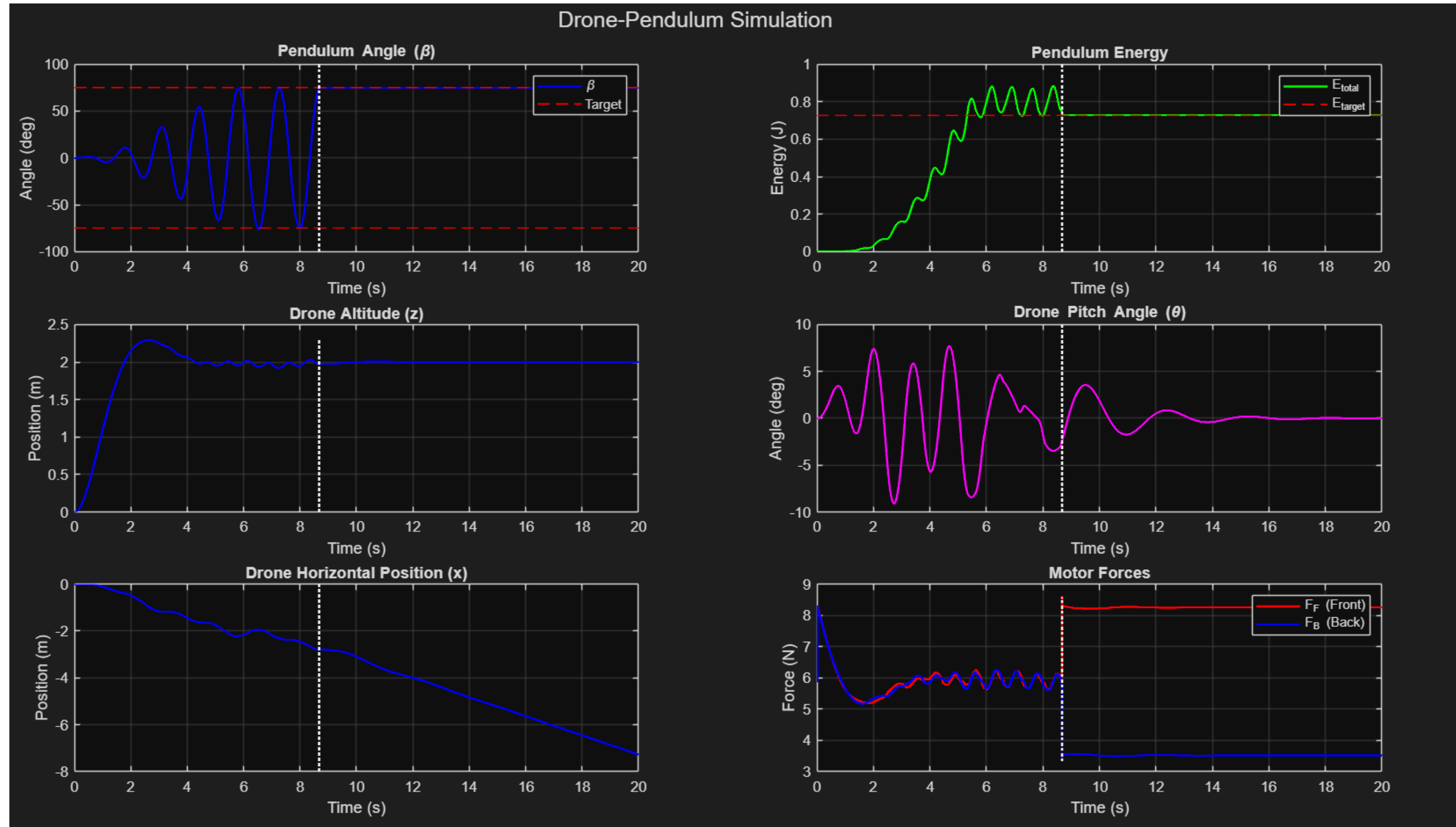
This law effectively "kicks" the pendulum in the direction of its motion.  $\dot{\theta} \propto -\dot{\beta}$  creates pseudo-forces that reinforces the swing.

# SIMULATION

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# RESULTS:



# RESULTS:

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- The controller successfully pumps energy, increasing oscillation amplitude from rest.
- At approx 8.6 s, the required energy is achieved, and the pendulum is locked.
- Despite the sudden shift in dynamics, the pitch angle stabilises to 0 within 2 seconds, preventing a crash.
- The controller automatically biases the front motors to carry the forward-projecting weight of the pendulum.
  - Front Motor ( $F_F$ ): Jumps to 8.2 N.
  - Back Motor ( $F_B$ ): Drops to approx 3.5 N.

# FUTURE SCOPE

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- Develop a control strategy that ensures stable convergence to a desired horizontal position (x-z plane), eliminating drift following the locking event
- Expand the dynamic model to incorporate yaw and lateral motion, enabling complete 6-DOF control during all operational phases.
- Develop and optimise ROS nodes for implementing swing-up, locking, and stabilisation controllers on the physical UAV platform.
- Develop the electromagnetic latch design for faster response, reliability, and reduced effect on mass distribution.
- Incorporate robust control methods to account for aerodynamic disturbances, cable flexibility, and actuator saturation limits

**Thank You**